

Dr. Mukesh Sharma

Research Scientist | 7+ Years in Sustainable Energy & Wastewater Treatment |
Expertise in Microbial Fuel Cells | Overseas Experience (UK, Korea, India)

Location: Bristol, United Kingdom

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Google Scholar: <https://scholar.google.co.in/citations?user=TGG8rQMAAAAJ&hl=en>

ResearchGate: https://www.researchgate.net/profile/Mukesh-Sharma-8?ev=hdr_xprf

PROFESSIONAL SUMMARY

Accomplished Research Fellow with a Ph.D. in Earth Science Technologies and extensive expertise in bio-electrochemical systems, microbial fuel cells (MFC), and sustainable agriculture. Currently leading the work package for the €7M Horizon Europe "Mi-Hy" (<https://www.mi-hy.eu>) project at UWE Bristol, synchronizing deliverables across an 8-partner international consortium. Proven track record in securing funding (ACORN, Royal Commission 1851 partnership), leading international collaborations, and bridging the gap between academic research and commercial application (MagTech, LettUs Grow). Expert in advanced material characterisation and the development of circular, self-sustaining hydroponic ecosystems.

PROFESSIONAL EXPERIENCE

Research Fellow (Microbial Hydroponics) University of the West of England (UWE), Bristol, UK | April 2024 – Present

Lead researcher for the UWE, Bristol for the Horizon Europe-funded "Mi-Hy" (<https://www.mi-hy.eu>) project, integrating Microbial Fuel Cell (MFC) technology with hydroponics to transform wastewater into resources.

Research & Technical Innovation:

- **Prosthetic Rhizosphere Development:** Conceptualised and engineered a "prosthetic rhizosphere" to enable natural nitrogen fixation in hydroponics, eliminating dependency on synthetic fertilisers.
- **Closed-Loop Ecosystems:** Established a circular hydroponic framework, successfully integrating MFC catholytes (sourced from University of Southampton) to replace inorganic nutrient feeds.

- **Algorithm Design:** Developed and implemented bespoke light-optimization algorithms to improve power-to-biomass ratios, allowing systems to operate efficiently on limited bio-electrochemical power.
- **Material Science Innovation:** Led the transition from plastic to sustainable materials by collaborating with KU Leuven to design and prototype "smart" ceramic hydroponic pots.
- **Data-Driven Phenotyping:** Collaborated with Sony CSL (Paris) to develop AI-driven phenotyping systems using image-based stress detection for real-time plant health monitoring.

Project Management & Leadership:

- **Work Package Leader:** Lead responsibility for the design and execution of Work Package, synchronising deliverables across an 8-partner international consortium including, Sony Computer Science Laboratory (CSL) Paris, CSIC (Spain) and BIOFACTION (Vienna).
- **Global Exhibition Management:** Managed the technical design and installation of "SPIKA," a 6-meter bio-hybrid tower exhibited at the **Milan Triennale**. Coordinated logistics and supervised PhD students from the University of Milan for the 9-month exhibition.
- **Mentorship:** Supervise PhD candidates, MSc students, and mentor researchers from the "Hop-On" partner (Faculty of Technology Novi Sad), aiding their integration into European collaborative frameworks.

Commercialisation & Grant Acquisition:

- **Grant Success:** Played a key role in securing the **ACORN grant**, leveraging consortium expertise to obtain bridging funds for research expansion.
- **Industrial Partnerships:** Established a Non-Disclosure Agreement (NDA) with **Mediterranean Agro Tech (MagTech)** to research hydroponic cotton.
- **Fellowship Proposals:** Authored a proposal for the Royal Commission for the Exhibition of 1851, securing **LettUs Grow** as a primary industrial sponsor.
- **Strategic Planning:** Developing Technology Readiness Levels (TRLs) and Technology Performance Levels (TPLs) for the European Innovation Council (EIC) portfolio to transition lab-scale technology to market-ready products.

Postdoctoral Fellow (NRF Project) Chungbuk National University, South Korea | March 2023 – February 2024

- Awarded the National Research Foundation (NRF) grant to lead a project on "Hydrogen Generation from wastewater using renewable sources of energy".

- Designed efficient electrode materials and developed microbial electrolysis technologies for hydrogen generation.
- Synthesised size-optimised graphene-based materials for nano-formulations in drug delivery.

Assistant Project Scientist Indian Institute of Technology (IIT) Guwahati, India | *December 2022 – February 2023*

- Developed bioelectrochemical technologies for simultaneous wastewater treatment and efficient hydrogen generation from waste sources.
- Supervised undergraduate and graduate students in electrocoagulation and membrane-based separation processes.

Senior Project Research Associate Indian Institute of Technology (IIT) Kharagpur, India | *August 2022 – October 2022*

- **Project:** "Development of Smart Portable Bioelectrochemical Toilet" (DST Funded).
- Synthesised polymeric and ceramic proton exchange membranes for large-scale prototypes handling human waste treatment and resource recovery.

Junior Project Assistant Indian Institute of Technology (IIT) Kharagpur, India | *August 2015 – July 2017*

- **Project:** "Potential of power production using salinity gradients from the Hooghly estuarine region" (INCOIS Funded).

EDUCATION

- **Ph.D. in Earth Science Technologies (CORAL)** | IIT Kharagpur, India | 2022
 - *Thesis:* Development of high-performance, cost-effective proton exchange membranes for simultaneous wastewater treatment and sustainable energy extraction.
 - *Fellowship:* MHRD PhD Fellowship, Govt. of India.
 - **Master (Research) in Earth Science Technologies** | IIT Kharagpur, India | 2018
 - *Grade:* 9.55/10 (INCOIS Scholarship).
 - **Bachelor of Engineering (Electronics & Communication)** | Bhabha Engineering and Research Institute, Bhopal, India | 2012
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PUBLICATIONS

Journal Articles

1. **Modesto L.R., Baquedano I., Ramalheite E.Ö., Mena S., Sharma M., et al.** (2026). Integrating electroactive microorganisms into active soil management strategies. *Frontiers in Microbiology*, 17:1753999. DOI: [10.3389/fmicb.2026.1753999](https://doi.org/10.3389/fmicb.2026.1753999)
2. **Mukesh Sharma**; Pramanik, A.; Bhowmick, G.D.; Tripathi, A.; Ghangrekar, M.M.; Pandey, C.; Kim, B.-S (2023). Premier, Progress, and Prospects in Renewable Hydrogen Generation: A Review. *Fermentation*, 9, 537. <https://doi.org/10.3390/fermentation9060537>
3. **Mukesh Sharma**, Piyal Mondal, A. D. Sontakke, A. Chakraborty & Mihir K. Purkait (2021a); High-performance graphene-oxide doped cellulose acetate-based ion exchange membrane for environmental remediation applications, **International Journal of Environmental Analytical Chemistry**, DOI: 10.1080/03067319.2021.1975276
4. **Mukesh Sharma**, Pranjal Pratim Das, Trishla Sood, A. Chakraborty & Mihir K. Purkait (2021b); Ameliorated polyvinylidene fluoride-based proton exchange membrane impregnated with graphene oxide, and cellulose acetate obtained from sugarcane bagasse for application in microbial fuel cell; **Journal of Environmental Chemical Engineering**, October 2021, Volume 9, Issue 6; Article 106681
5. **Mukesh Sharma**, Pranjal Pratim Das, Trishla Sood, A. Chakraborty & Mihir K. Purkait (2022b); Reduced graphene oxide incorporated polyvinylidene fluoride/cellulose acetate proton exchange membrane for energy extraction using microbial fuel cells; **Journal of Electroanalytical Chemistry**, November 2021, Volume 907; Article 115890
6. **Mukesh Sharma**, Pranjal Pratim Das, A. Chakraborty & Mihir K. Purkait (2022a); Clean energy from salinity gradients using pressure retarded osmosis and reverse electrodialysis: A review; **Sustainable Energy Technologies and Assessments**, November 2021, Volume 49; Article 101687
7. **Mukesh Sharma**, Mondal, P., Chakraborty, A., Kuttippurath, J., Purkait, M.; Effect of different molecular weight polyethylene glycol on flat sheet cellulose membranes for evaluating power density performance in pressure retarded osmosis study. Published in **Journal of Water Process Engineering** (<https://doi.org/10.1016/j.jwpe.2018.05.011>).
8. **Mukesh Sharma**, Chakraborty A, K Jayanarayanan, Yadav AK (2018b) Potential Power Production from Salinity Gradient at the Hooghly Estuary System. **Innov. Ener. Res.** 7: 210. (doi: 10.4172/25761463.1000210)
9. Pranjal Pratim Das, **Mukesh Sharma**, & Mihir K. Purkait; Recent progress on electrocoagulation process for wastewater treatment: A review; **Separation and Purification Technology**. Volume 292, 1 July 2022, 121058; <https://doi.org/10.1016/j.seppur.2022.121058>.
10. Jadhav, D. A., Yu, Z., Hussien, M., Kim, J.-H., Liu, W., Eisa, T., **Sharma, M.**, Vinayak, V., Jang, J.-K., Wilberforce Awotwe, T., Wang, A., & Chae, K.-J. (2024). Paradigm shift in Nutrient-Energy-Water centered sustainable wastewater treatment system

- through synergy of bioelectrochemical system and anaerobic digestion. **Bioresource Technology**, 396, 130404. <https://doi.org/10.1016/j.biortech.2024.130404>
11. Singh, A., Kafle, S. R., **Sharma, M.**, & Kim, B. S. (2023). Comprehensive Review on Multifaceted Carbon Dot Nanocatalysts: Sources and Energy Applications. **Catalysts**, 13(11), 1446. <https://doi.org/10.3390/catal13111446>
 12. Khaled AbouAitah, Yingjie Bu, **Mukesh Sharma**, Beom Soo Kim, Preparation, characterization, and in vitro release studies of multifunctional nanoformulations designed by functionalized graphene nanosheets with natural compounds, **FlatChem**, Volume 50, 2025, 100824, ISSN 2452-2627, <https://doi.org/10.1016/j.flatc.2025.100824>.
 13. Cabulong, R. B., Kafle, S. R., Singh, A., **Sharma, M.**, & Kim, B. S. (2024). Biological production of nicotinamide mononucleotide: a review. **Critical Reviews in Biotechnology**, 1–18. <https://doi.org/10.1080/07388551.2024.2433993>
 14. **Mukesh Sharma**, Saroj Raj Kafle, Anju Singh, Arun Chakraborty, Beom Soo Kim (2024). **ChemCatChem**; Carbon quantum dots for efficient hydrogen production: A critical review <https://doi.org/10.1002/cctc.202400056>

Books (ELSEVIER Publications)

1. Blue Energy Extraction Using Salinity Gradients: A Critical Evaluation of Case Studies
2. Real-Life Applications on Electrocoagulation-Based Treatment of Water and Wastewater

Book Chapters

1. Simultaneous treatment of industrial wastewater and resource recovery using microbial fuel cell; In book: Resource Recovery in Industrial Waste Waters; DOI: [10.1016/B978-0-323-95327-6.00002-6](https://doi.org/10.1016/B978-0-323-95327-6.00002-6); **ELSEVIER (2023)**.
2. Extraction of clean energy from industrial wastewater using bioelectrochemical process; In the book: Resource Recovery in Industrial Waste Waters; DOI: [10.1016/B978-0-323-95327-6.00003-8](https://doi.org/10.1016/B978-0-323-95327-6.00003-8); **ELSEVIER (2023)**.
3. Recycling of water treatment plant sludge and its utilisation for wastewater treatment; In book: Resource Recovery in Drinking Water Treatment; DOI: [10.1016/B978-0-323-99344-9.00010-4](https://doi.org/10.1016/B978-0-323-99344-9.00010-4) **ELSEVIER (2023)**.
4. Energy storage properties of nanomaterials; **ELSEVIER (2021)**.
5. Polyurethane Foams as Packing and Insulating Materials; “**American Chemical Society**” (2022)
6. Biochar application for treatment of industrial effluent as a green technology: Opportunities, challenges, and future perspectives; **CRC PRESS (2021)**.
7. Restoration of contaminated agricultural soils by modified biochar; **CRC PRESS (2021)**.

8. Use of biochar for the remediation of heavy metal contaminated soil; **CRC PRESS** (2021).
9. Application of biochar in the removal of organic and inorganic contaminants from wastewater; **CRC PRESS** (2021).
10. Removal of the pharmaceutical compounds from wastewater by designer biochar; **CRC PRESS** (2021).
11. Recent advances in dye removal technologies by designer biochar; **CRC PRESS** (2021).
12. Purkait, M. K., Das, P. P., & **Sharma, M.** (2024a). A real-life application of the treatment of contaminated drinking water by a continuous electrocoagulation process. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 191–202). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00012-7>
13. Purkait, M. K., Das, P. P., & **Sharma, M.** (2024b). Advances in the electrocoagulation process: Fundamentals and mechanisms. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 69–91). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00013-9>
14. Purkait, M. K., Das, P. P., & Sharma, M. (2024c). Continuous electrocoagulation treatment of textile industry effluent located in Istanbul, Turkey. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 157–174). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00014-0>
15. Purkait, M. K., Das, P. P., & **Sharma, M.** (2024d). Environmental assessment of sludge generated from water treatment plants. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 237–257). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00006-1>
16. Purkait, M. K., Das, P. P., & **Sharma, M.** (2024e). Fluoride and iron mitigation in drinking water by the electrocoagulation process. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 203–219). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00010-3>
17. Purkait, M. K., Das, P. P., & **Sharma, M.** (2024f). Fundamentals of electrochemistry in water treatment. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 47–67). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00004-8>
18. Purkait, M. K., Das, P. P., & **Sharma, M.** (2024g). Hybrid ozonation–electrocoagulation treatment of biological oxidation treated wastewater from the steel industry, India. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 113–136). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00008-5>
19. Purkait, M. K., Das, P. P., & **Sharma, M.** (2024h). Integrated electrocoagulation–microfiltration process for the treatment of nanofiltration rejected effluent from the steel industry, India. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 175–189). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00009-7>
20. Purkait, M. K., Das, P. P., & **Sharma, M.** (2024i). Performance of the electrocoagulation process for the removal of arsenic from drinking water. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 221–236). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00003-6>

21. Purkait, M. K., Das, P. P., & **Sharma, M.** (2024j). Water and wastewater: Resource management and quality standards. In *Electrocoagulation Based Treatment of Water and Wastewater* (pp. 1–18). Elsevier. <https://doi.org/10.1016/B978-0-443-13892-8.00005-X>
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CONFERENCES & PRESENTATIONS

- **Sharma M., et al.** (2021). International Chemical Engineering Conference "100 Glorious Years of Chemical Engineering & Technology," Dr. B R Ambedkar NIT, Jalandhar, India.
 - **Sharma M., et al.** (2021). International Conference on Advances in Chemical, Biological, and Environmental Engineering, MNIT Jaipur, India.
 - **Sharma M., et al.** (2020). WATER 2020, IIT Guwahati, Assam, India.
 - **Sharma M., et al.** (n.d.). INDO-EU joint workshop on "The Recent Developments in Microbial Fuel Cell and Membrane Bioreactor Technology," IIT Kharagpur, India.
 - **Sharma M., et al.** (n.d.). International conference on 21st-century energy needs, IIT Kharagpur, India.
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MEDIA COVERAGE

International & Project Coverage (UWE Bristol / Europe)

- **UWE Research Blog:** [UWE Bristol Researchers Developing a Prosthetic Microbiome for ‘Vertical Farms’](#)
- **Triennale Milano Exhibit:** [Mi-Hy brings a 6-meter artistic prototype to Triennale Milano](#)
- **Future Tech Week 2025:** [Mi-Hy Feature](#)
- **EIC-SUNERGY Roadmapping:** [Mi-Hy at the joint EIC–SUNERGY Event](#)
- **Deepsync:** [Mi-Hy featured on Deepsync](#)
- **Bilateral Workshop (UAB):** [Projects Mi-Hy and CONFETI at the UAB](#)
- **EIC Portfolio:** [First Annual Meeting & Press Release](#)
- **Workshops:** [Microbial Fuel Cell Workshop at Southampton Labs](#) | [System Integration Workshop in Paris](#) | [Microbial Hydroponics Event at CIVA, Brussels](#)

National Media Coverage (India - Wastewater to Energy Technology)

- **Department of Science & Technology (Govt. of India):** [Video Coverage 1](#) | [Video Coverage 2](#)

- **Invited Lecture:** Sustainable Energy Generation (YouTube)
- **The Hindu;** URL: <https://www.thehindu.com/news/national/other-states/from-iit-guwahati-a-device-to-turn-wastewater-into-energy/article65482952.ece>
- **The Times of India;** URL: <https://timesofindia.indiatimes.com/city/guwahati/iit-g-develops-device-to-create-green-energy-from-waste-water/articleshow/91950306.cms>
- **The Sentinel;** URL: <https://www.sentinelassam.com/topheadlines/iit-guwahati-develops-tech-to-generate-green-energy-by-treating-wastewater-594642>
- **Gadgets Now;** URL: <https://www.gadgetsnow.com/tech-news/iit-guwahati-researchers-develop-technology-to-generate-green-energy-by-treating-wastewater/articleshow/91952823.cms>
- **The Economic Times;** URL: <https://economictimes.indiatimes.com/industry/renewables/iit-guwahati-researchers-develop-tech-to-generate-green-energy-from-waste-water/articleshow/91942126.cms>
- **Republic World;** URL: <https://www.republicworld.com/education/news/iit-guwahati-researchers-develop-tech-to-generate-green-energy-from-waste-water-articleshow.html>
- **The Deccan Herald;** URL: <https://www.deccanherald.com/national/iit-guwahati-researcher-develops-device-to-produce-green-energy-from-waste-water-1114386.html>
- **India Today;** URL: <https://www.indiatoday.in/education-today/news/story/iit-guwahati-develops-technology-to-generate-energy-by-treating-wastewater-1956946-2022-06-01>
- **The Telegraph India;** URL: <https://www.telegraphindia.com/edugraph/news/iit-guwahati-to-generate-energy-by-treating-wastewater/cid/1867870>
- **The Print;** URL: <https://theprint.in/india/iit-guwahati-researchers-develop-tech-to-generate-green-energy-from-waste-water/979358/>
- **The Week;** URL: <https://www.theweek.in/wire-updates/national/2022/06/01/des28-iit-research-waste-water.html>
- **Etv Bharat;** URL: <https://www.etvbharat.com/assamese/assam/state/kamrup-metropolitan/iit-guwahati-develops-tech-to-generate-green-energy-by-treating-wastewater/assam20220602112933631631165>
- **The Sentinel Assam;** URL: <https://hindi.sentinelassam.com/guwahati-city/iit-guwahati-develops-tech-to-generate-green-energy-by-treating-wastewater-594719>
- **India Tv;** URL: <https://www.india.com/technology/iit-guwahati-develops-bio-electrochemical-device-to-generate-green-energy-by-treating-wastewater-heres-how-5427271/>
- **Amar Ujala;** URL: <https://www.amarujala.com/india-news/researchers-of-iit-guwahati-develop-device-to-convert-waste-water-into-energy>
- **Guwahati Plus;** URL: <https://guwahatipius.com/guwahati/iit-guwahati-develops-technology-to-generate-energy-by-treating-wastewater>
- **North East Today;** URL: <https://www.northeasttoday.in/2022/06/01/iit-guwahati-researchers-develop-bio-electrochemical-device-to-generate-sustainable-energy-from-waste-water/>

- **The Meghalayan;** URL: <https://themeghalayan.com/iit-guwahati-develops-technology-to-generate-energy-by-treating-wastewater/>
- **The Shillong Times;** URL: <https://theshillongtimes.com/2022/06/01/iit-g-develops-technology-to-generate-energy-by-treating-wastewater/>
- **The Arunachal Times;** URL: <https://arunachaltimes.in/index.php/2022/06/02/iit-guwahati-researchers-develop-tech-to-generate-green-energy-from-waste-water/>
- **Gujarat Samachar;** URL: <https://www.gujaratsamachar.com/news/national/energy-from-waste-water-iit-guwahati-invented-a-transformer-machine>
- **KRC Times;** URL: <https://www.krctimes.com/news/iit-guwahati-develops-technology-to-generate-energy-by-treating-wastewater/>
- **Eastern Mirror Nagaland;** URL: <https://easternmirrornagaland.com/iit-g-develops-tech-to-generate-green-energy-by-treating-wastewater/>
- **Bhaskar;** URL: <https://www.bhaskarlive.in/iit-guwahati-develops-tech-to-generate-green-energy-by-treating-wastewater/>
- **Andhram;** URL: <https://www.andhram.com/national/iit-guwahati-develops-tech-to-generate-green-energy-by-treating-wastewater/>
- **The Assam Tribune;** URL: <https://assamtribune.com/guwahati/iit-guwahati-develops-tech-to-generate-green-energy-by-treating-wastewater-1372323>
- **Education Today;** URL: <https://educationtoday.co/blogs/?p=18313>
- *Samachar, KRC Times, Eastern Mirror Nagaland, Bhaskar, Andhram, The Assam Tribune, Education Today.*

TECHNICAL SKILLS

Instrumentation:

- Surface area and pore analyzer (Quantachrome Autosorb iQ)
- Thermo-gravimetric Analysis (TGA, NETZSCH instrument, TG 209 F1 Libra®, Germany)
- Raman spectroscopy (Horiba jobin-Vyon T6400, Lab Raman HR)
- Powder X-ray diffractometer (Bruker D8 Advanced XRD measurement system, Japan)
- Fourier Transform Infrared Spectroscopy (FTIR, Shimadzu, IR Affinity 1, Japan)
- Field Emission Scanning Electron Microscopy (FESEM) (Zeiss, Sigma 1430 VP)
- Transmission Electron Microscopy (TEM) (JEM 2100, JEOL)
- Energy Dispersive X-ray (EDX) facility
- Atomic Force Microscopy (AFM) (Agilent, 5500 series)
- Ultraviolet-visible Spectroscopy (UV-Vis) (Shimadzu UV-1601 PCspectrophotometer)
- Fluorescence Spectrophotometer (Make:PerkinElmer, Singapore, Model: LS 45)

Software: Origin, Chemdraw, Microsoft Excel.

Data Skills: Database creation, Critical thinking, Data visualization, Data analysis.

REFERENCES

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